www.tntextbooks.in

As a traveller you would have got a chance to observe the plants. It is an interesting fact that all plants are not unique. Just spend some time to listen to nature. You can notice plants with attractive leaves, flowers and fruits.

Can you say all plants are healthy and uniform in growth? Some plants are not healthy and show symptoms like texture changes, stunted growth, chlorosis, necrosis and so on. Can you tell what is the reason for all these symptoms? It may be due to infection of microbial pathogens or climatic factors or due to mineral deficiency.

In this chapter we are going to learn about classification of minerals, their functions, deficiency diseases and symptoms, nitrogen metabolism and special modes of nutrition. Further, how can these ideas help us to improve productivity in agriculture?

Plants naturally obtain nutrients from atmosphere, water and soil. Carbon, hydrogen and oxygen are called as skeletal elements and constitute about 94% of dry weight. These elements play an important role in the formation of organic compounds such as carbohydrates, fats and protein. These non-mineral elements are obtained from air and water. Minerals are classified based on essentiality. **Arnon** and **Stout** (1939) gave criteria required for essential minerals:

1. Elements necessary for growth and development.
2. They should have direct role in the metabolism of the plant.
3. It cannot be replaced by other elements.
4. Deficiency makes the plants impossible to complete their vegetative and reproductive phase.

12.1 Classification of minerals

12.1.1 Classification of minerals based on their quantity

Essential elements are classified as **Macronutrients, Micronutrients** and **Unclassified minerals** based on their requirements. Essential minerals which are required in higher concentration are called Macronutrients. Essential minerals which are required in less concentration called are as Micronutrients.

Minerals like Sodium, Silicon, Cobalt and Selenium are not included in the list

Historical events in mineral nutrition

Van Helmont (1648) – made first observation of mineral nutrition, noticed over a period of 5 years soil lost only 56 g in nourishing a seedling into tree. Increase in organic substance comes from water alone.

Wood word (1699) – Soil provides mineral nutrients required for their growth.

De Saussure (1804) – plant growth depends on nitrogen and other elements absorbed by roots from soil.

Liebig (1840) – gave the "law of minimum" which states that productivity of soil depends on amount of essential elements present in minimum quantity.

Julius Von Sachs (1860) – Demonstrated growing a plant in a defined nutrient solution.

William Frederick Goerick (1940) – Gave the term Hydroponics and commercial technique.